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विज्ञान और प्रौद्योगिकी विभाग

National Institute of Geo-Informatics
Science & Technology
Survey of India
Department of Science & Technology

STANDARD OPERATING PROCEDURE
for
Field Operation of
NRTK Method using RTX
(Trimble R980 GNSS & Trimble Access)

Course Name	500 (Surveying Engineer)			
Training Module Name	NRTK Method using RTX (Trimble R980 & Trimble Access)			
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1. Introduction

This Standard Operating Procedure (SOP) provides detailed instructions for conducting NRTK (Network Real Time Kinematic) surveys using the Trimble R980 GNSS Receiver with Trimble Access software using the RTX service of Trimble. It defines the workflow from equipment setup, job creation, GNSS correction configuration, measurement operations and data export, ensuring standardized high-accuracy GNSS field practices.

2. Objective

The objectives of NRTK surveying using Trimble R980 with RTX are:

- To establish GNSS control points with centimeter accuracy without base stations.
- To execute topographic, engineering, and stakeout surveys efficiently.
- To utilize IMU tilt compensation to increase productivity.
- To ensure standardized, accurate, and reliable GNSS data collection.

3. Note on RTX

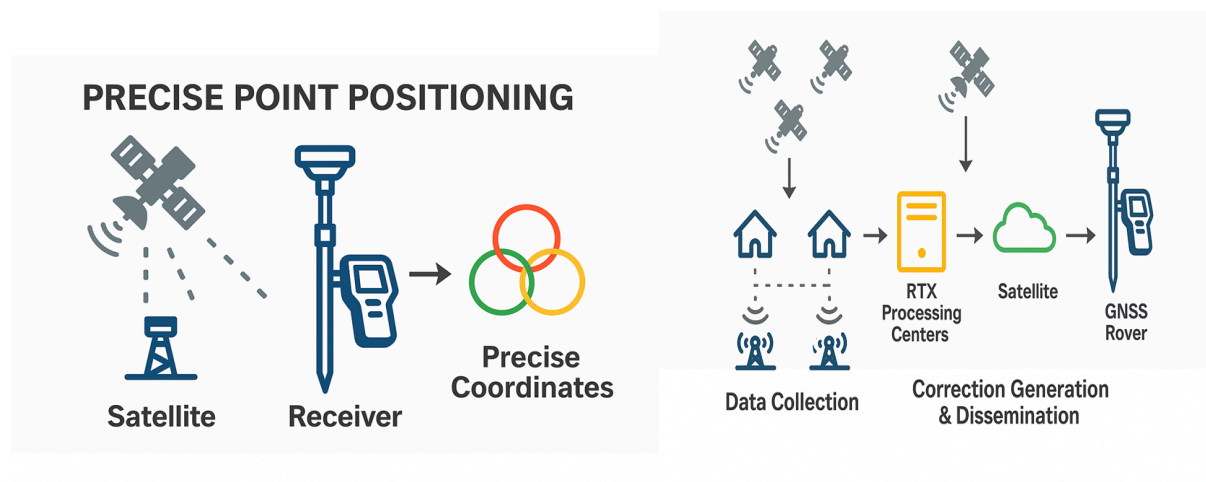
Trimble RTX (Real-Time eXtended) is a global GNSS correction service providing 2 cm accuracy without a local base station or VRS network. RTX uses a worldwide monitoring network, satellite orbit/clock modeling, and atmospheric correction algorithms. It delivers corrections through L-band satellites or the internet (IP).

Key features:

Global coverage 2 cm horizontal accuracy Convergence: <1 min in RTX Fast regions; 1–3 min globally. Works without base station xFill Premium maintains accuracy during interruptions **RTX is ideal for remote areas, highways, long corridors, and projects without RTK/CORS infrastructure or Places without Internet connectivity.**

The Trimble RTX service works in two ways.

- a. If the place of Survey does not have Internet connectivity or if CORS network is absent the RTX (SV) (SV-Satellite Vehicle) is used- in this the corrections are directly transmitted via satellite to the receiver to calculate the position by PPP (Precise Point Positioning)**
- b. If the place of survey has internet but no CORS network then RTX IP (Internet Protocol) is used. In this method the corrections are sent via the internet so that the receiver can calculate its position.**



4. Hardware & Software

4.1 Hardware

- Trimble R980 GNSS Receiver
- Data Controller (TSC5/TSC7/Getac/Laptop)
- Fixed 2m survey pole
- Li-ion batteries and charger
- SIM card/Internet Connection/Hotspot (for IP RTX)
- USB or wireless data transfer module



4.2 Software

- Trimble Access (v2024 or later)
- Trimble RTX subscription (CenterPoint RTX included for 1 year)
- Optional: Trimble Business Center (TBC) Version 2025.11



5. Pre requisites

- Basic knowledge of GNSS and coordinate systems
- Active internet (IP RTX)
- Clear sky visibility
- Fully charged equipment

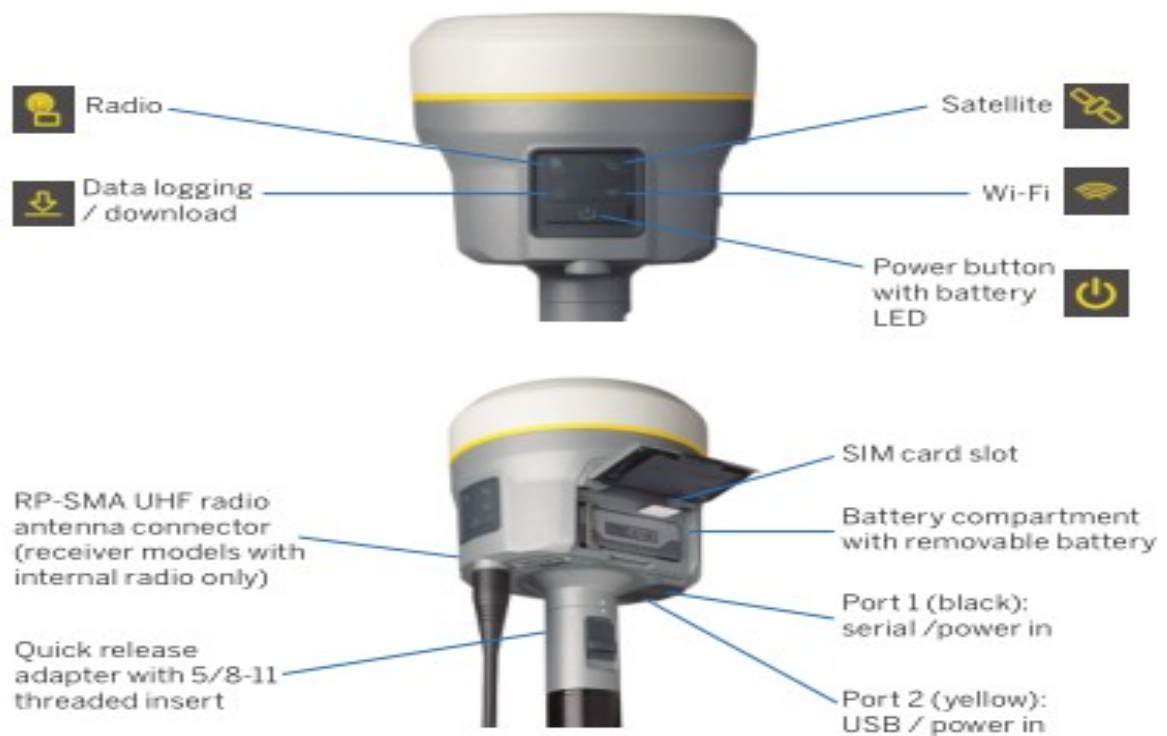
6. Roles & Responsibilities

Surveyor shall:

- Configure job, coordinate system, and survey style
- Validate accuracy on known point
- Collect measurements as per required tolerance
- Export and secure data

7. Specifications of Trimble R980 GNSS Receiver

- GNSS Constellations: GPS, GLONASS, Galileo, BeiDou, QZSS, NAVIC L5
- Accuracy with RTX: 2 cm horizontal, 5 cm vertical
- Internal modem: 4G LTE with 2G fallback
- Internal memory: 9 GB
- Dual-band 450/900 MHz UHF radio (region dependent)
- IMU/Tilt Compensation: 5 mm + 0.4mm/degree tilt
- RTX Convergence: <1 min (RTX Fast), 1–3 min global
- Includes 1-year CenterPoint RTX subscription



8.Setting up the GNSS Equipment

8.1 Setting up Survey Pole

- Use fixed 2m pole for consistency
- Ensure pole tip is firmly placed on ground

8.2 Mounting the R980

- Attach R980 securely to quick-release adapter
- Ensure locking mechanism is firm

8.3 Leveling & IMU Alignment

Connect R980 to controller

- Enable eBubble
- Move/rotate pole until IMU alignment shows green tick

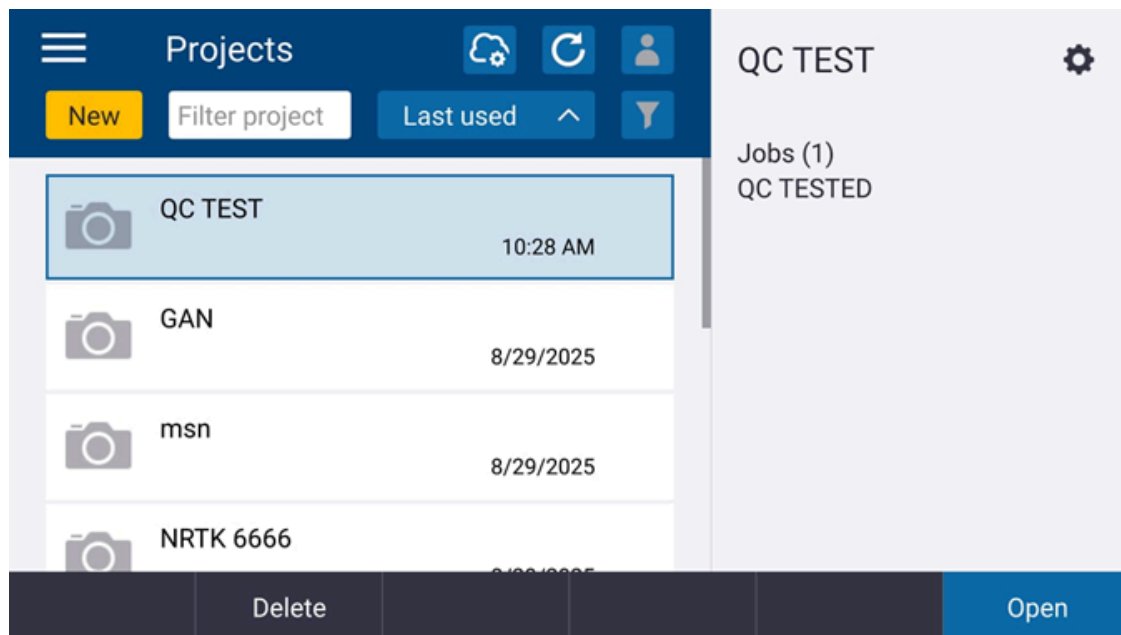
8.4 Centering on Control Point

- Used for site calibration or accuracy check
- Center pole exactly using eBubble

9. Creating a Job in Trimble Access

9.1 Create New Project & Job

- Open Trimble Access in Controller TSC5 or Getac, in menu Create New Project and inside New Project create New Job



- Give Project Name e.g. "NRTK TESTING" and click >>Enter

The screenshot shows the 'New project' form in Trimble Access. The form has a 'Properties' tab and fields for Name, Description, Reference, Location, and Image. The 'Name' field is filled with 'NRTK TESTING'. The other fields are empty with a question mark. At the bottom, there are buttons for 'Esc', a camera icon, and 'Enter'.

- Inside project create job, Give a name to this new job e.g. "NRTK Test"

← New job: NRTK TESTING/NRTK Test ☆

☒ Create from template ☐ Create from JobXML or DC file

Job name: NRTK Test

Template: Metric Scale Only

Properties

Coord. sys.: Scale: 1.0000000000

Units (Dist.): Meters

Layer manager: None

Feature library: None

Esc Accept

9.2 Set Coordinate System

- Click on Scale
- Click on " Select from library"

☰ Select coordinate system

Select coordinate system

☒ Select from library

☐ Key in parameters

☐ No projection / no datum

☐ Broadcast RTCM

Esc Next

- In the Select coordinate system
- Select System: World wide/UTM from drop down menu
- Select Zone: Calculate and select from dropdown menu as per your working area Longitude (44N for Hyderabad)

- Select Local datum: WGS 1984 (7P) from drop down options
- Global reference datum: WGS 1984 (Default)
- Global reference epoch: 2005(Default)
- Displacement model: ITRF2020 (Default)
- Tectonic plate: Detect automatically
- Use Geoid model Click YES (if any local Geoid model is available) & select appropriate GGF file or else select NO.

Select coordinate system

System

World wide/UTM

Zone

44 North

Local datum

WGS 1984 (7P)

Global reference datum

WGS 1984

Global reference epoch

2005.00

Displacement model

ITRF2020

Tectonic plate

(Detect automatically)

Use geoid model

☐ No

Use datum grid

Coordinates

Esc

Key in

Store

- Use datum grid: No
- Coordinates: Grid.
- Project Height: Give average Height of the working area.
- Click >>Store

Select coordinate system

Global reference datum

WGS 1984

Global reference epoch

2005.00

Displacement model

ITRF2020

Tectonic plate

(Detect automatically)

Use geoid model

☐ No

Use datum grid

No

Coordinates

Grid

Project height

490

Esc

Key in

Enter

- Click Additional settings

← New job: NRTK TESTING/NRTK Test ☆

Properties	
Coord. sys.	44 North (World wide/UTM)
Units (Dist.)	Meters
Layer manager	None
Feature library	None
Cogo settings	Ground
Additional settings	Off
Media file	Previous point
Reference	?

Esc Accept

- Enable Add to csv & give name e.g."NRTK Testing" to .csv file

☰ Additional settings

Feature library

Use attributes of base code

☒

Add to CSV file

Enable ☒ Yes

CSV file name

Point name range for job

Apply point name range

☐ No

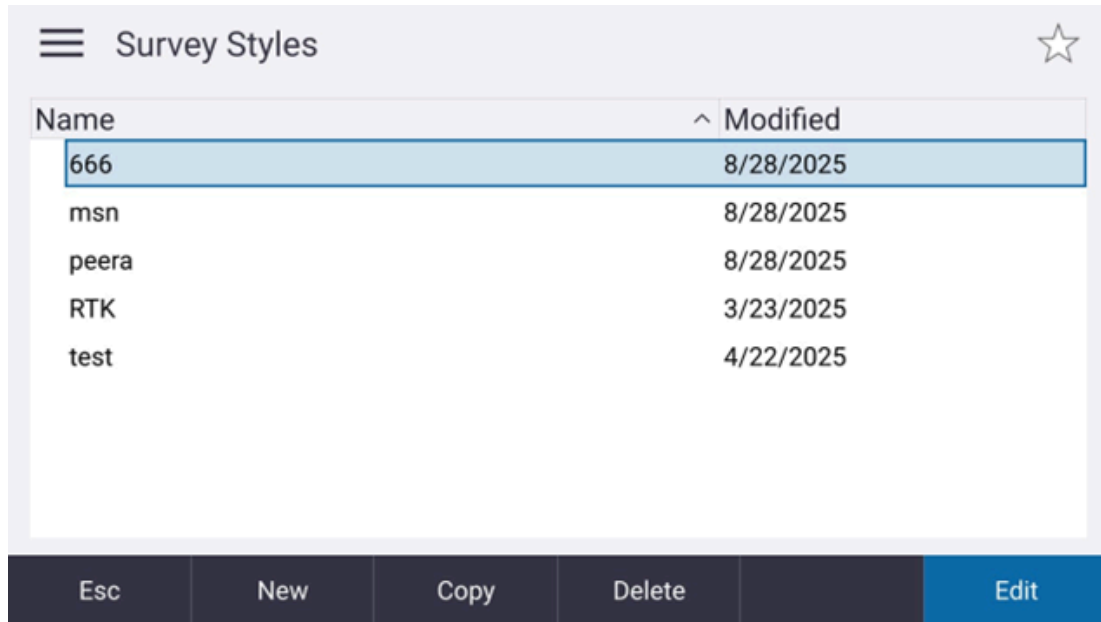
Esc Enter

- Click Enter

Now a New Project & a new job with a defined projection system has been created.

10. Creating NRTK Survey Style using RTX (SV)

- Now we have to create a New Survey style for NRTK method, for that click main menu
- Go to settings

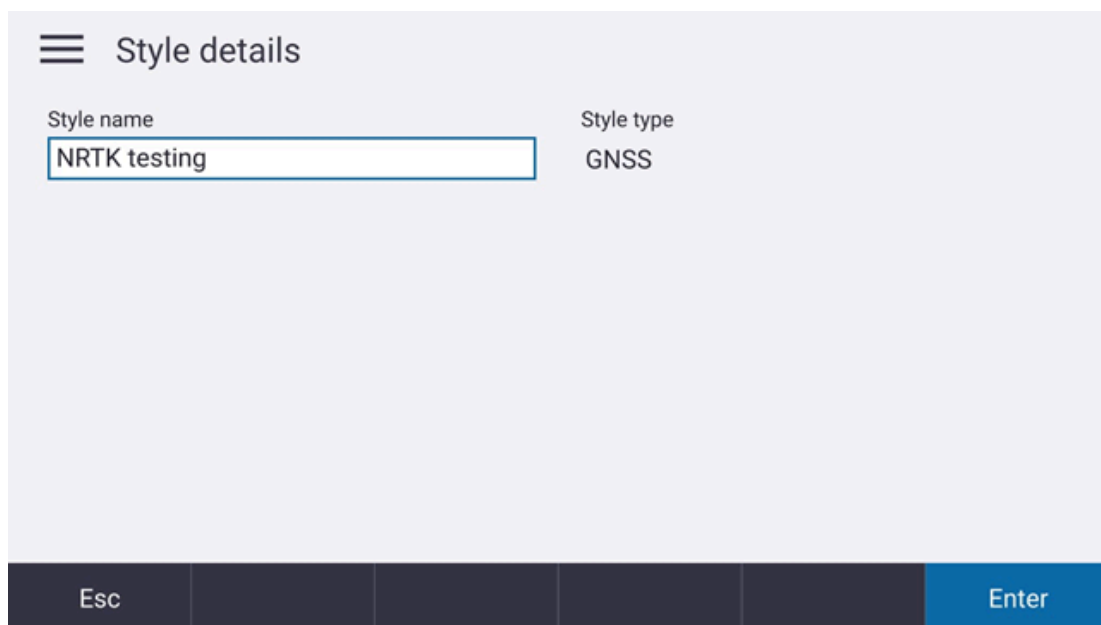


The screenshot shows a 'Survey Styles' menu with a list of styles. The 'Name' column is on the left and the 'Modified' column is on the right. The style '666' is highlighted. The bottom navigation bar has buttons for 'Esc', 'New', 'Copy', 'Delete', and 'Edit'.

Name	Modified
666	8/28/2025
msn	8/28/2025
peera	8/28/2025
RTK	3/23/2025
test	4/22/2025

Esc New Copy Delete Edit

- Under Survey Styles Click New to create a new Survey style
- Give Style name e.g. "NRTK Testing" , - & click >>Enter



The screenshot shows the 'Style details' form. The 'Style name' field is labeled 'NRTK testing' and the 'Style type' field is labeled 'GNSS'. The bottom navigation bar has buttons for 'Esc' and 'Enter'.

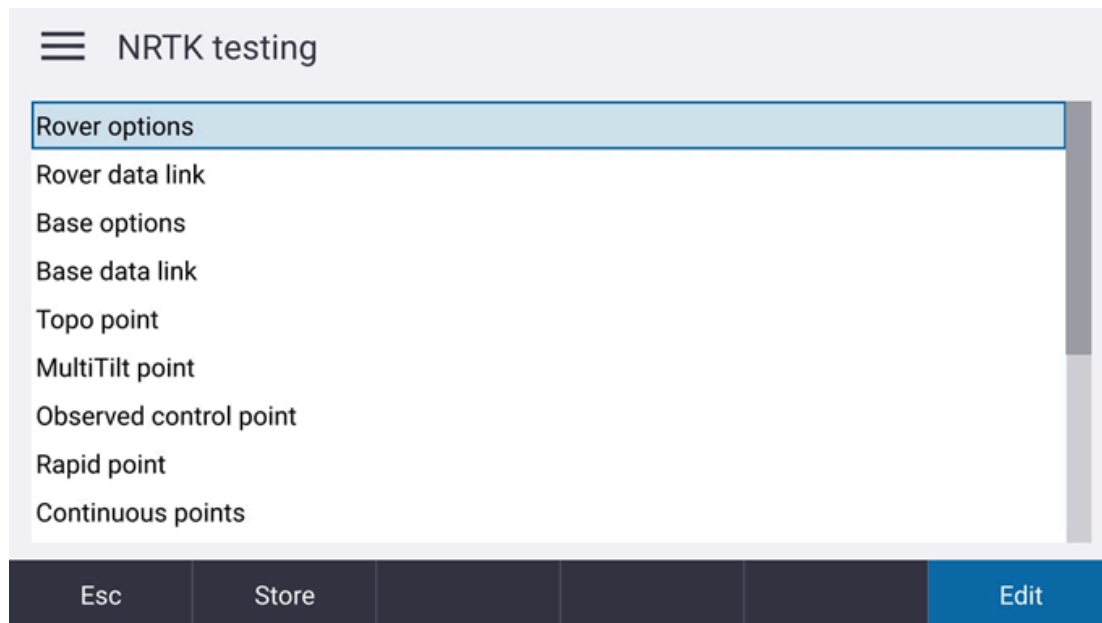
Style name: NRTK testing

Style type: GNSS

Esc Enter

10.1 Rover Options

- Next Select Rover options new style



- In Rover Options
- Select Survey Type as RTK
- Antenna: R980 Internal
- Broadcast Format: RTX SV
- Antenna Height 2m (if you are using bi-pod pole), if using a Tripod enter exact height after measuring with tape
- Measured to: Bottom of Quick Release
- Part Number: Default
- Serial Number: You can enter Serial number of receiver or by default it will automatically take after connecting with the GNSS receiver.
- In Tilt option: Check on IMU tilt compensation & E bubble functions
- AR viewer: check off (This option is not available in TSC5 controller)

Augmented Reality (AR) feature used for visualizing survey data in the field It Uses the controller's camera (e.g., TSC5, TDC600, or tablet) together with GNSS position from the R980.

☰ Rover options

Survey type
RTK

Broadcast format
RTX (SV)

Antenna

Type
R980 Internal

Antenna height
2.000m

Serial number
7953

Measured to
Bottom of quick release

Part number
128221-XX

Tilt

IMU tilt compensation
☒

eBubble functions
☒

Elevation mask

Esc Accept

- Store points as: Positions
- Elevation mask: 10° to 15° (Preferably 10°)
- PDOP Mask: <6

☰ Rover options

AR viewer
☒

Store points as
Positions

Elevation mask
10°

PDOP mask
6.0

GNSS Signal Tracking

GPS
☒

Use L2e
Yes

Esc Accept

- GNSS Signal Tracking: Select all

Rover options

GNSS Signal Tracking

GPS

☒

GPS L2C

☒

GLONASS

☒

QZSS

☒

NavIC

☒

Use L2e

Yes

L5

☒

Galileo

☒

BeiDou

☒

Esc

Accept

- In Roving precision
- Auto tolerance: No
- Horizontal tolerance (DRMS): 0.015m.
- Vertical tolerance: 0.030m.
- xFill: check on

Trimble Access

Rover options

GPS

☒

GPS L2C

☒

GLONASS

☒

QZSS

☒

Use L2e

Yes

L5

☒

Galileo

☒

BeiDou

☒

Roving precision

Auto tolerance

☒ No

Horizontal tolerance (DRMS)

0.015m

Vertical tolerance (1 sigma)

0.030m

11. Field operation using RTX(SV)

To perform survey operation using RTX(SV) only Rover settings need to be done.

No internet is required since all the corrections will be received via satellite.

To start Survey first connect/pair the R980 GNSS receiver to the Getec of the CS5 controller via bluetooth. (Refer SOP of NRTK Survey using R8s)

On connection the screen will appear as below.

The battery indicators of R980 GNSS receiver & Controller will be displayed.

The number of Satellites will also be displayed.

Goto>> Measure>>Select the Survey Style

Select the style created

- Now select >>Measure points for measuring as a point

Measure code for (for Topo features with some code)

Topo point: To measure a point in less time i.e.2sec

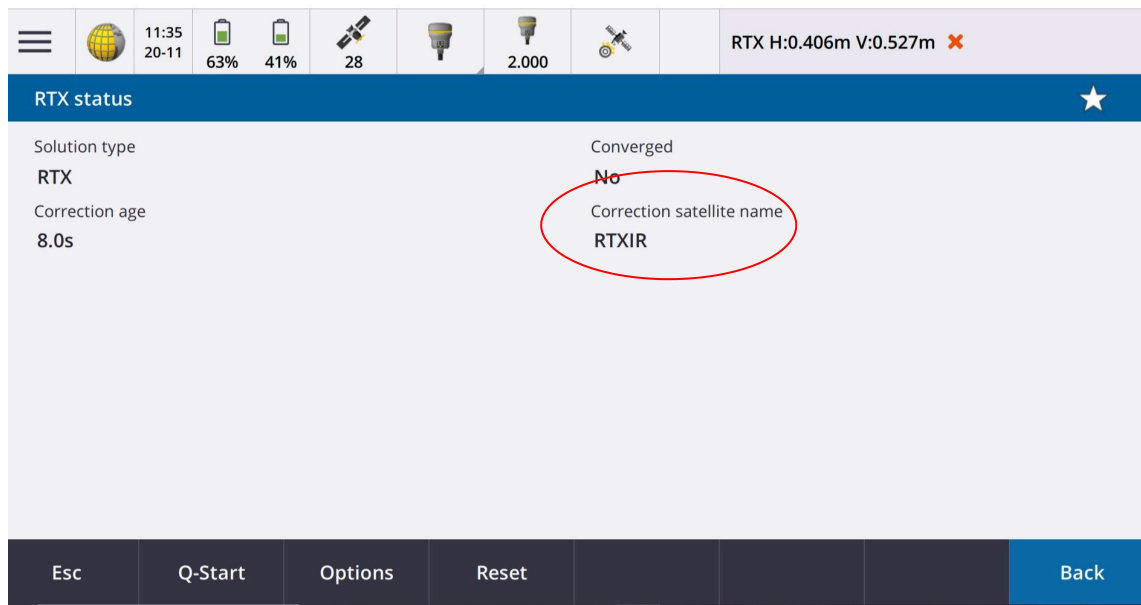
Measure to surface(option is used when you want to measure a point but record its elevation (Z value) relative to a reference surface)

Continuous topo (It is a mode used for topographic surveys where you want to quickly capture a large number of points along the ground surface continuously rather than stopping at each point.

Site Calibration (This method Site calibration (also called Localization) is the process of aligning GNSS coordinates (WGS84 / ITRF / UTM) with your local survey control system)

RTX- Satellite Vehicle is active.

On clicking this satellite Icon following screen is visible. Here the RTX satellite name- RTXIR is shown.



In the List of available satellites the Satellite RTX is also shown. Here it is noticeable that only 1 frequency i.e. L1 is being transmitted by this Satellite.

Trimble Access

RTX H:0.311m V:0.459m ✖

Satellites

SV	Az	Elev	L1 CA/...	L2E/L...	L2C/C...	L5/Alt...
✓ C39	34°	31°	44.5	46.0	42.7	43.3
✓ C40	126°	23°▲	34.7	37.2	34.2	34.2
✓ C41	339°	51°▲	41.2	46.7	40.2	35.2
✓ C42	132°	16°▼	29.0	34.5	26.5	----
C59	99°	19°	31.0	----	----	----
C60	175°	69°▼	40.5	----	----	44.5
RTX	164°	69°	43.3	----	----	----
✓ J2	69°	26°▲	40.5	----	45.2	47.0
J7	105°	32°	38.7	----	47.5	47.7
J8	---	---	----	----	----	----

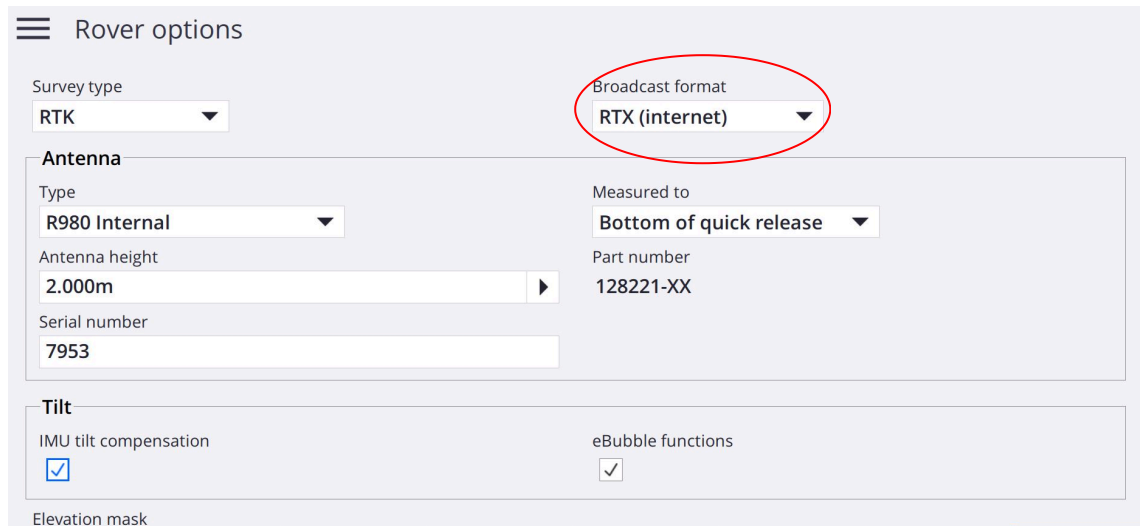
The 'RTX' row is circled in red. At the bottom, there is a navigation bar with buttons: Esc, Info, Reset, Options, and Plot.

If we see the PLOT view we can see the RTX satellite.

12. Rover settings (for RTX IP)

If using RTX IP i.e. using RTX with Internet, In Rover settings only change 1 setting.

In Broadcast format choose RTX(internet)

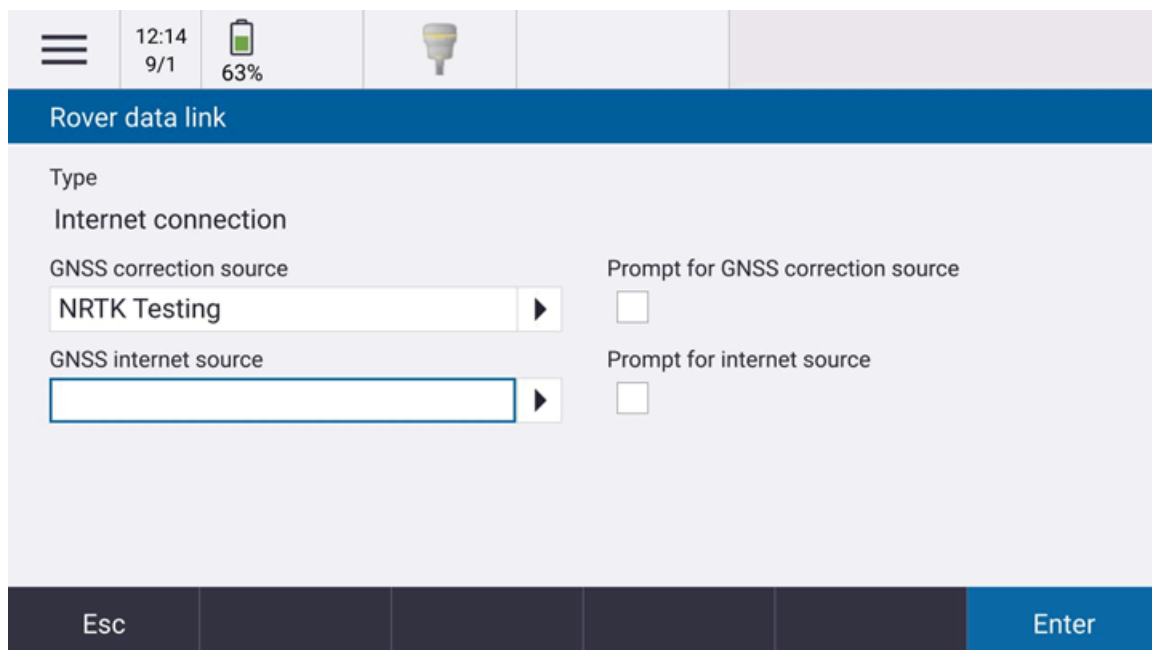


The screenshot shows the 'Rover options' menu. The 'Broadcast format' dropdown is highlighted with a red circle and set to 'RTX (internet)'. Other settings include: Survey type: RTK; Antenna Type: R980 Internal; Antenna height: 2.000m; Serial number: 7953; Measured to: Bottom of quick release; Part number: 128221-XX; IMU tilt compensation: checked; eBubble functions: checked; and Elevation mask.

Rest of the settings would be exactly the same as mentioned in 10.1 above

13. Rover Data Link Settings (for RTX IP)

- Next click Rover data Link



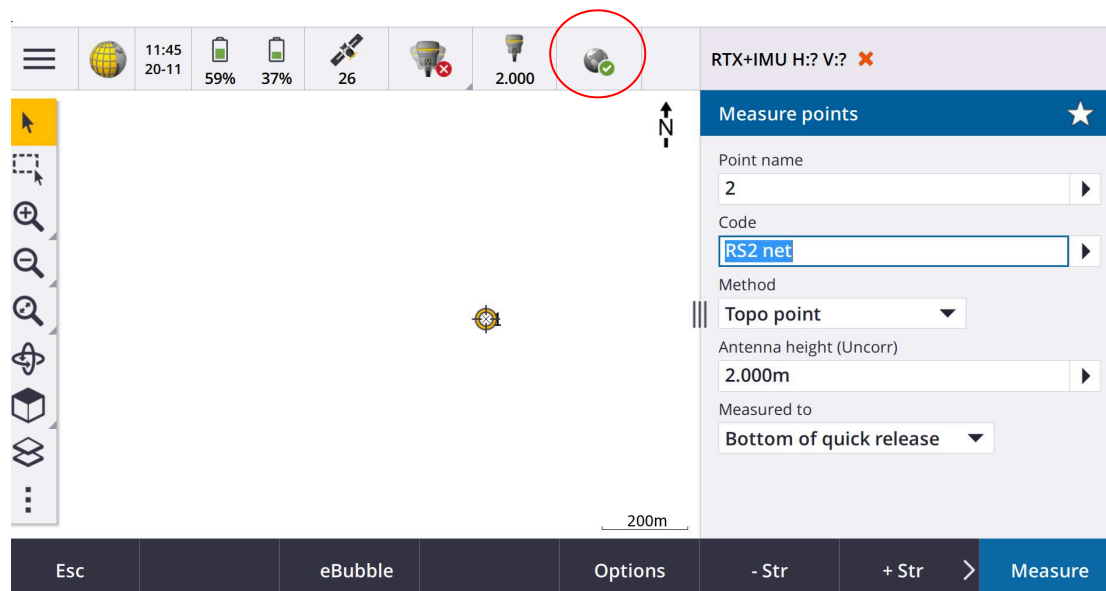
The screenshot shows the 'Rover data link' settings screen. The 'Type' is set to 'Internet connection'. The 'GNSS correction source' is set to 'NRTK Testing'. The 'GNSS internet source' is empty. There are checkboxes for 'Prompt for GNSS correction source' and 'Prompt for internet source', both of which are unchecked. The bottom of the screen shows 'Esc' and 'Enter' buttons.

- Click the arrow in front of GNSS internet Source
- Give any name such RTX net
- In Mountpoint name select RTXIP from the drop down menu
- Check on Use RTX(Internet)
- Provide internet to the controller via hotspot or SIM (as applicable)

14. Field operation using RTX(IP)

Goto>> Measure>>Select the Survey Style

Here the icon of a *Spinning Globe with Green check box* appears when it is connected to RTX IP

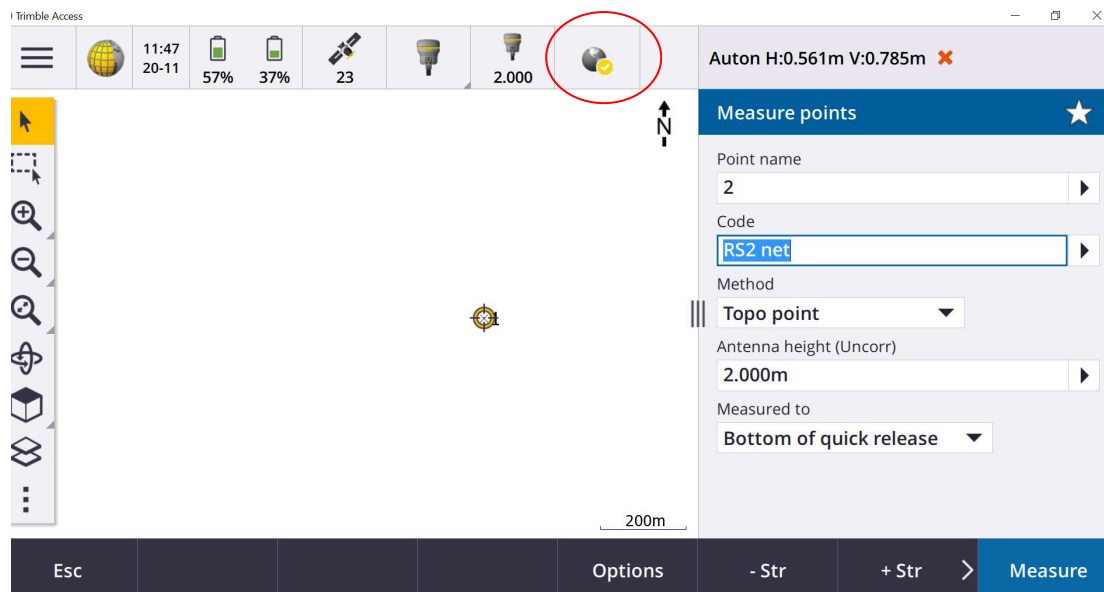


To do measurements: Enter point Name, Code, Method, Antenna Height & Measured to.

As soon as the Horizontal & Vertical Accuracy is achieved (within given tolerances)

The red cross will change to green tick. Here you can store the point.

If connectivity is lost the *Spinning Globe with Yellow check box* icon appears.



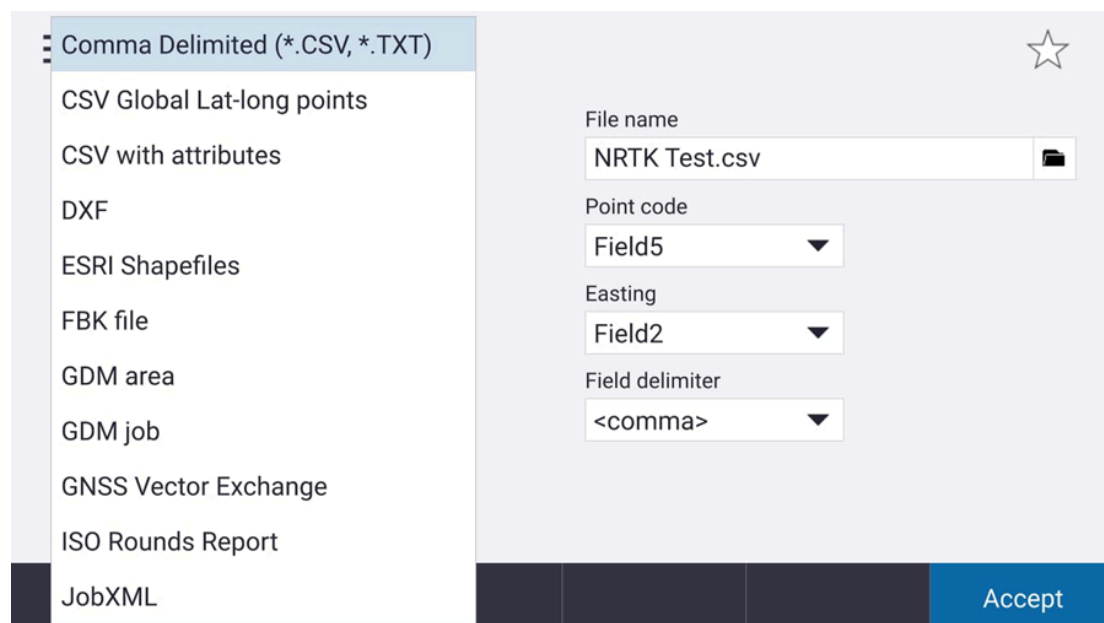
15. Downloading of Data

Exporting the job:

After Ending the survey, exit Trimble Access and

- Open the Project
- Select the job under project

At the bottom you will find Export option>>click



You will find different options to export the job, Select Comma Delimited (*.CSV, *.TXT)

For exporting in .csv file, or any other option as required.

File name .csv file

- Click Accept

Export

File format: Comma Delimited (*.CSV, *.TXT) ▼

File name: NRTK Test.csv

Point name: Field1 ▼

Point code: Field5 ▼

Northing: Field3 ▼

Easting: Field2 ▼

Elevation: Field4 ▼

Field delimiter: <comma> ▼

View created file: ☒

Esc Accept

In Select points

- Select All points option
- Click Accept

Select points

Enter single point name

Select from list

All grid points

All keyed in points

Points within radius

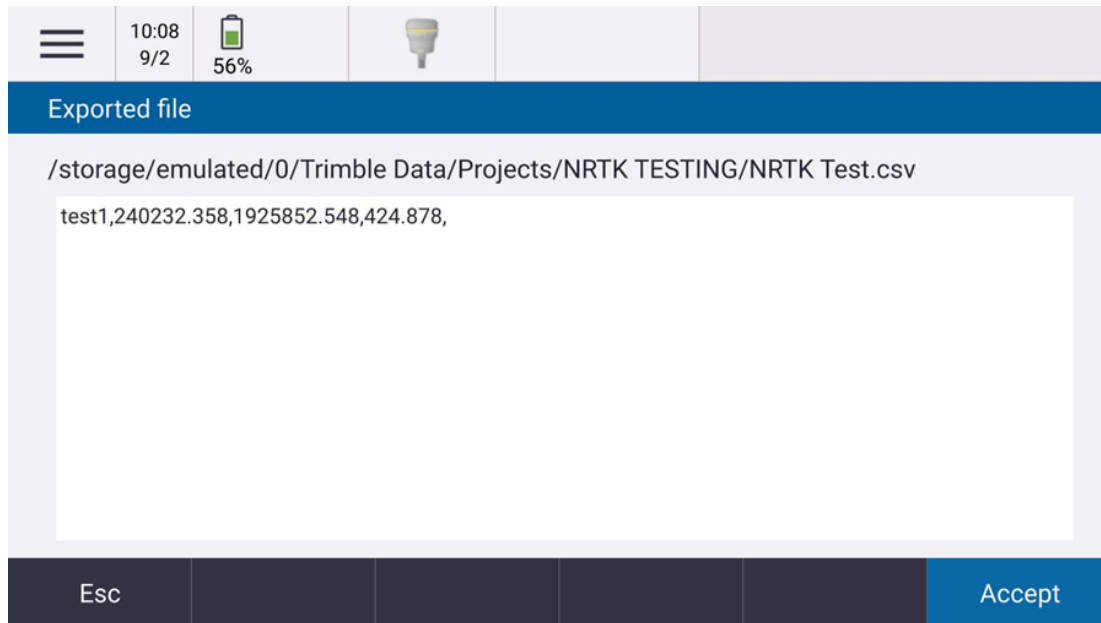
All points

Points with same code

Points by name range

Esc Accept

You will find the path in which the output file is saved, go to that path and select the concerned file and copy to a pen drive or share through Whatsapp.



16. Conclusion

RTX(SV) is very useful in areas having no internet connectivity or having no CORS network.

RTX(IP) is very useful in areas having no CORS network.